# CHANDU PRASAD BHAIRAPU



## AI Engineer

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## ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in AI-first engineering with coding agents, from agentic coding workflows with Claude Code and critical review of AI-written code to API and cloud integrations on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by close work with non-technical stakeholders and a strong foundation in Python development.

## EXPERIENCE

### Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

**Kniit GmbH**

November 2025 – present Muldestausee, Germany

- Worked AI-first as an AI-augmented engineer with Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- Sole developer owning the full knowledge-extraction pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, letting domain experts validate and approve knowledge-graph changes.
- **Tech stack: Claude Code, Python, LangGraph, LangChain, LiteLLM, MCP, RAG, Neo4j, MongoDB, Postgres, FastAPI, React, Docker, Kubernetes, Azure**

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### GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

**Covestro Deutschland AG**

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows and turn them into clear requirements, then designed and deployed an agentic LangGraph application that routes each request to vector database queries, general inference or summarization.
- Developed a React frontend and a Python backend with FastAPI APIs, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

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### GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

**Covestro Deutschland AG**

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, CDK), Docker, Streamlit**

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### Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

**Covestro Deutschland AG**

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) for the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

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### Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

**Cognizant Technology Solutions**

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time series analysis of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

## PROJECTS

**Project: Spec-Driven Coding Agent Harness | Claude Code | MCP | GitHub Actions | Feature Flags**

- Spec-first harness: a feature request becomes a spec with edge cases and pytest acceptance tests; Claude Code implements it under skills, an MCP server for spec context and hooks that block secrets.
- GitHub Actions ships each change behind a feature flag with monitoring of error rates, automatic rollbacks, structured logging and an agent eval suite.

## TECHNICAL SKILLSET

- **AI-first Engineering:** Claude Code (custom skills, agentic coding workflows), AI coding agents, AI agents (multi-agent systems), prompt engineering, LangGraph, LangChain, LiteLLM, MCP, RAG, evaluation against ground truth
- **Software:** Python, FastAPI, REST and external APIs, Python microservices, React (beginner), Postman, clean code, Git, GitLab CI/CD, Agile
- **Cloud & Data:** AWS (Lambda, SQS, DynamoDB, EC2, S3, CDK), Azure, Docker, Kubernetes, Postgres, SQL, MongoDB, Neo4j, vector databases
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

## EDUCATION

**Master of Science in Artificial Intelligence** — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

**Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering)** — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India